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The online platform for area-wide management of Western X-disease

Extension Program Summary

Target Audience

The management of insect-vectored plant diseases often requires a combination of cultural, biological, and chemical controls. In cases where the vector is highly mobile and feeds on a wide range of host plants, successful management often requires intervention that is coordinated on a regional scale (Chandler and Faust 1998, Faust 2008). Sweet cherry growers on the west coast face a similar problem where the leafhoppers carry Western X-disease pathogens through orchards at an exceptionally high rate. Leafhoppers acquire disease pathogens by feeding on infected trees. Cherry growers apply insecticides to control the population of vectors, however, uncoordinated application of insecticides leave refuge areas in the growing regions. Plus, growers tend to delay the removal of infected trees which exacerbate the secondary infection by leaving the source of disease pathogen in the orchard. Thus, Western X-disease management calls for an area-wide management program. The target audience of this extension program are Western-X disease management decision makers in the sweet cherry growing regions of Washington, California, Idaho, and Oregon. This includes those that have direct input into disease management planning such as owners, managers, commercial disease consultants, and extension officers.

Development of extension program

There are several successful area-wide management (AWM) programs in agriculture and natural resource management in the United States, including the General Beale Pilot Project which is a case study for the use of AWM in controlling Pierce's disease in grapevines (Sisterson and Stenger 2013), the Ecological Areawide Management of Leafy Spurge, and the Hawaii Fruit Fly Areawide Pest Management Program (ArcGIS StoryMaps 2021). The common challenging issue faced by AWM programs, even successful programs, is uncertainty about the commitment of participants which can determine the future of the program. For example, the AWM program that

was established in 2010 to limit the spread of huanglongbing disease in citrus orchards of Florida but eventually failed due to lack/little involvement of grower (Singerman, Lence, and Useche 2017; Singerman and Useche 2019; Lence and Singerman 2023). Thus, adding a tool to decrease uncertainty about commitment of participants and showcase the farm-level economic impact of AWM is a significant step to increase the participation in AWM programs. The AWM of leafhoppers aims to limit the spread of Western X-disease through coordinated application of insecticides and removal of infected trees in the sweet cherry growing regions of Washington state. The bottleneck of the program is that growers are facing the dilemma of sacrificing immediate revenues, by removing infected trees, to prevent potential future disease spread. The issue becomes more consequential when considering growers' uncertainty about other participants' willingness to remove infected trees.

The program proposes a web-based platform to reduce the uncertainty surrounding the effectiveness of AWM and commitment of participants in real time. The program contributions will be threefold. First, the web-based platform provides recommendation on the optimal tree removal rates based on the number of participants in the program, disease pressure in the area, age distribution of trees in the area, and other important similar information by using a bioeconomic model. Second, the commitment of each participant is shared with other participants and the public. The participants' commitment to the AWM program can be measured by the share of removed trees in the orchard which can be judged based on the satellite imagery and remote sensing surveys. Third, the platform emphasizes the importance of timely removal of infected trees and showcases the potential positive impacts of AWM at farm-level by comparing economic and ecological measures from participating and non-participating areas.

The program will be implemented in phases. In the first phase, growers will get to know the AWM web-based platform. We will include a how-to video to instruct potential platform users how to use the tool. This video will be built in collaboration with applied economists, entomologists, and plant pathologists from Washington State University College of Agriculture, Human, and Natural Resources (WSU-CAHNRS) Extension, and video communication experts. In the second phase, when growers are familiar with the platform, after watching the video, they will proceed to input information. The platforms will allow growers to input their orchard characteristics incorporating assumptions for disease pressure in the area. In the third phase, the platform tool will present potential action plans, expected profits for each action plan, and

recommendations based on the best action plans. In the fourth phase, the platform will provide insights about the impact of AWM on growers' profits. The platform tool will summarize the impact of individual actions and map the potential added value of AWM commitment in a way that will strengthen the strategic decisions made by growers.

The advantage of this platform is that the decision-making and assumptions will be specific to each grower, which provides an immediate view of the future impacts over the life span of orchard. For example, a grower can evaluate the impact of removing 10% of symptomatic trees and compare the expected benefits with the break-even point.

Platform Preview

We collaborate closely with applied economists, entomologists, and plant pathologists at WSU-CAHRNS extension. This is to show our commitment to tailor the platform to sweet cherry growers needs in Washington state while maintaining its broad capabilities. Collaboration with PREC allows us to refine, and adjust, the platform. The final aim of the AWM platform is to provide solutions for strategic decision making. Figure 1 below illustrates a preview of the first phase of platform using historical data for a representative grower.

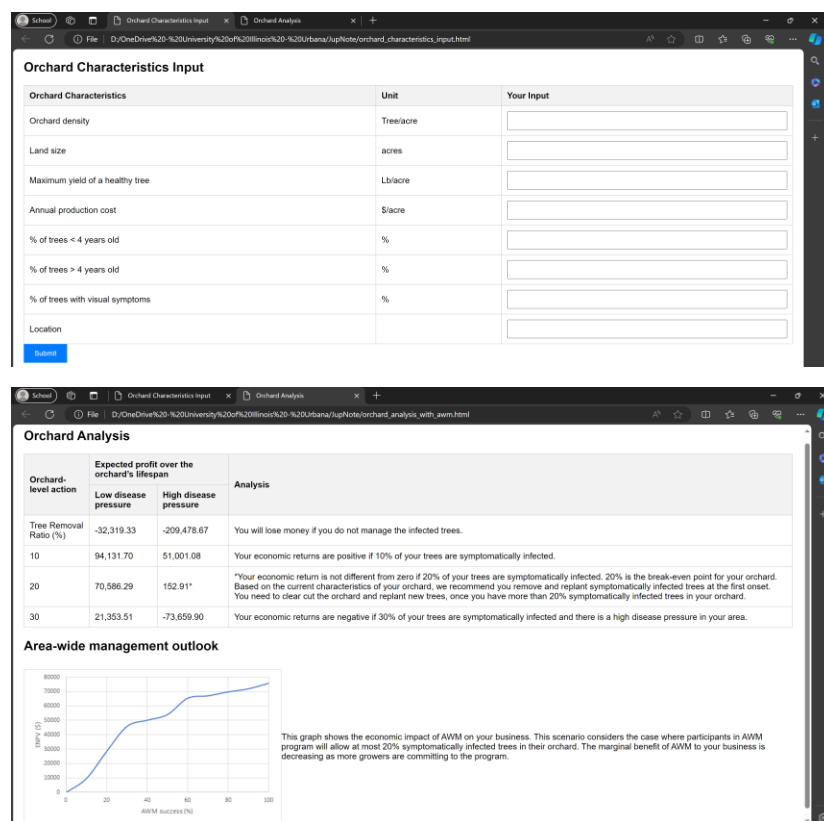


Figure 1.

The goals and objectives of the extension program are as follows:

1. Educate and demonstrate how cherry growers can use a web-based platform to benefit from leafhopper AWM program.
2. Provide resources and materials that aid users of the platform.
3. Ensure that the platform is applicable and useable for AWM of other agricultural diseases across the U.S.

Team Members' Responsibilities, Delivery Plans and Communication Methods

Other than me, the extension program team consists of:

- Dr. R. Karina Gallardo, Professor at School of Economic Sciences, Washington State University
- Dr. Scott Harper, Associate Professor, Virology Director, CPCNW, Washington State University

We plan to work on the alpha version of the platform for the remainder of the year. The alpha version platform will be launched early 2025 to get feedback from stakeholders such as Northwest Cherry Grower Association, cherry growers, and disease management consultants who have different scale and scope from WSU-CAHNRS extension faculty. We plan to incorporate feedback from the initial launch and refine the platform for the remainder of 2025. The beta launch is set to be launched in early 2026. The extension program will begin with annual in-person training sessions. Upon completion of each cohorts' training, attendees will have access to online resources such as videos, exercises, and fact sheets. Our initial scope is to deliver the program to cherry growers in the Yakima valley of Washington and then offer it state-wide.

The platform will be made available on the WSU-CAHNRS extension website. As mentioned, an online training portal will accompany the platform. The portal will consist of a series of recorded videos that demonstrate how to use the platform and exercises illustrating scenarios that are likely to be faced by a cherry grower. In addition, an annual training symposium at WSU will also be held, and periodic fact sheets with updates of the platform will also be made available. Lastly, staff will be available to answer questions on the phone and e-mail.

References

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- Singerman, Ariel, Sergio H Lence, and Pilar Useche. 2017. "Is Area-Wide Pest Management Useful? The Case of Citrus Greening." *Applied Economic Perspectives and Policy* 39 (4): 609–34. <https://doi.org/10.1093/aepp/ppx030>.
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Evaluation Plan¹

The evaluation plan aims to assess the effectiveness, usability, and impact of the AWM web-based platform. Regarding the web-based platform, we will conduct an evaluation survey to people who have used the tool to get their feedback on what is working and what is not working, and how we could improve the tool. Plus, this survey allows us to assess the initial level of knowledge. A follow-up survey will be conducted after one year of which further can be used to inform the support team about the shortcomings of the platform and gauge the understanding of growers about the AWM program. In addition, we prepare a short quarterly report for presenting at growers' meeting which further increases the communication between growers and AWM support team. Also, we plan to prepare content suitable for social media (i.e., Facebook, Instagram, X, and YouTube) to increase outreach. Moreover, we regularly publish our findings in popular articles like the Good Fruit Grower, American Fruit Grower, and universities' extension webpages.

The evaluation will be conducted in accord with the implementation phases of the program.

Evaluation objectives are as follows.

- Determine the extent to which the AWM platform assists growers in making informed decisions about tree removal rates, based on economic and ecological factors.
- Assess the usability of the platform, including ease of navigation, inputting data, and understanding recommendations.
- Measure the commitment of participants to the AWM program and the impact of shared commitment on program effectiveness.
- Evaluate the economic and ecological impact of AWM at the farm level.

Evaluation Phases:

Phase 1: Introduction to the Platform

Surveys: Conduct surveys among growers to assess their initial perceptions and understanding of the AWM platform.

User Feedback: Gather feedback from growers regarding the clarity and usefulness of the how-to video and initial platform interface.

¹ Logic model is attached separately.

Expert Review: Engage with experts from WSU-CAHNRS Extension to review the content and effectiveness of the instructional video.

Phase 2: Data Input and Familiarization

Usage Analytics: Track usage metrics to understand how growers interact with the platform, including the frequency of data input and time spent on each section.

User Interviews: Conduct interviews with growers to gather qualitative feedback on their experience with inputting data and navigating the platform.

Error Analysis: Monitor for any errors or inconsistencies in data input to identify potential usability issues or misunderstandings.

Phase 3: Action Plans and Recommendations

Case Studies: Select a sample of growers to participate in case studies where their actual decisions are compared to recommendations provided by the platform.

Post-Decision Surveys: After receiving recommendations, gather feedback from growers to assess the perceived usefulness and accuracy of the platform's recommendations.

Comparative Analysis: Analyze the economic and ecological implications of actions taken based on platform recommendations compared to traditional methods.

Phase 4: Impact Assessment

Economic Analysis: Quantify the economic impact of AWM on participating growers by comparing profits before and after implementing recommendations from the platform.

Ecological Monitoring: Utilize satellite imagery and remote sensing surveys to assess changes in tree health and disease prevalence in participating orchards. Also, tracking the participants' commitment to the program by comparing the GIS maps of participating orchards before and after the program.

Participant Surveys: Conduct surveys to measure growers' perceptions of the impact of AWM on their profits and ecological sustainability.

Data Collection and Analysis:

Data collected from surveys, interviews, and usage analytics will be analyzed using both quantitative and qualitative methods. Statistical analysis will be employed to identify correlations between platform usage, grower decisions, and program outcomes. Comparative analysis will be conducted to assess the differences between participating and non-participating areas in terms of economic performance and ecological indicators.

Reporting and Dissemination:

Findings will be summarized in a comprehensive evaluation report detailing the effectiveness and impact of the AWM platform. Recommendations for program improvement and future iterations of the platform will be provided based on evaluation results. Results will be disseminated through presentations, publications, and outreach efforts targeting stakeholders in the agriculture community.

Timeline:

The evaluation will be conducted iteratively alongside the implementation phases of the program, with data collection occurring at each phase and analysis conducted periodically to inform program adjustments and improvements.

Resources:

- Evaluation team comprising agricultural economists, technology experts, and program stakeholders.
- Access to usage analytics tools, survey platforms, and data analysis software.
- Collaboration with WSU-CAHNRS Extension for expert input and program feedback.

Background and Research

Khashi is a third year Ph.D. student in the department of Agricultural and Consumer Economics (ACE) at the University of Illinois at Urbana-Champaign (UIUC). His advisor is Dr. Shady S. Atallah. His research interests focus on resource economics and bioeconomic, as well as on adoption of sustainable disease management practices in specialty crops. After earning his Ph.D., he hopes to obtain a faculty position with an Extension appointment. He obtained his M.Sc. degree from University of New Hampshire in May 2021, with his thesis focusing on studying the role of agroforestry in mitigating the impact of climate change on coffee production in Colombia.

Conference Presentations

Ghorbai, K., (2024), “*A Game-Theoretic Mechanism to Combat Antibiotic Resistance in Animal Agriculture*”, The Agricultural & Applied Economics Association, New Orleans, USA.

Ghorbai, K. and Atallah, S., Scott Harper., R. Karina Gallardo (2024), “*Estimating the Value of Area Wide Disease Management: A Bioeconomic Model of Western X-Disease in Cherry Orchards*”, The Agricultural & Applied Economics Association, New Orleans, USA.

Ghorbai, K. Atallah, S., (2024), “*Optimizing Antibiotic Use in Plant Agriculture: A Decision Model under a Potential Antibiotic Ban*”, World Conference on Natural Resource Modeling, New Haven, USA.

Ghorbani, K., and Atallah, S. (2021), “*Estimating the Value of Disease Regulation Services Under Climate Change: A Bioeconomic Model of Coffee Leaf Rust and Shade-grown Coffee*”, The Agricultural & Applied Economics Association, Austin, USA.

Mentor's Information

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Role: Mentor and assisting in the design and execution of the extension program.